

Problem 3

The rotational frequency of a body is $\nu = 100$ Hz.

1. What is the speed of the body if it rotates along a circular path with a radius $r = 1$ m?
2. What is the normal acceleration of the body?
3. What angular deceleration is required to stop the body after completing three revolutions?
4. What is the magnitude of the total acceleration of the body immediately after the deceleration starts?

Solution

Part (a) - Speed of the Body

The speed of a body moving along a circular path is related to the angular velocity ω by the formula:

$$v = \omega r.$$

The angular velocity ω is related to the rotational frequency ν as:

$$\omega = 2\pi\nu.$$

Substituting the given values:

$$\omega = 2\pi \times 100 = 200\pi \text{ rad/s.}$$

The linear speed is:

$$v = 200\pi \times 1 = 200\pi \approx 628.32 \text{ m/s.}$$

Part (b) - Normal Acceleration

The normal (centripetal) acceleration is given by:

$$a_n = \frac{v^2}{r}.$$

Substituting $v = 200\pi$ and $r = 1$:

$$a_n = \frac{(200\pi)^2}{1} = (200\pi)^2 = 40000\pi^2 \approx 394784.18 \text{ m/s}^2.$$

Part (c) - Angular Deceleration

The angular displacement during deceleration is $\theta = 3$ revolutions, or in radians:

$$\theta = 3 \times 2\pi = 6\pi \text{ rad.}$$

The final angular velocity is zero, and the initial angular velocity is $\omega_0 = 200\pi$. Using the kinematic equation for angular motion:

$$\omega^2 = \omega_0^2 + 2\alpha\theta.$$

Since $\omega = 0$:

$$0 = (200\pi)^2 + 2\alpha(6\pi).$$

Solving for α :

$$\alpha = -\frac{(200\pi)^2}{2 \times 6\pi} = -\frac{40000\pi^2}{12\pi} = -\frac{40000\pi}{12} \approx -10472.98 \text{ rad/s}^2.$$

Part (d) - Magnitude of Total Acceleration Immediately After Braking Starts

The total acceleration is the vector sum of the normal acceleration and the tangential acceleration due to angular deceleration. The tangential acceleration is:

$$a_t = r|\alpha|.$$

Substituting the values:

$$a_t = 1 \times 10472.98 = 10472.98 \text{ m/s}^2.$$

The magnitude of the total acceleration is:

$$a = \sqrt{a_n^2 + a_t^2}.$$

Substituting $a_n \approx 394784.18$ and $a_t \approx 10472.98$:

$$a = \sqrt{(394784.18)^2 + (10472.98)^2} \approx 395168.89 \text{ m/s}^2.$$

Final Answer

- (a) Speed of the body: $v \approx 628.32 \text{ m/s}$.
- (b) Normal acceleration: $a_n \approx 394784.18 \text{ m/s}^2$.
- (c) Angular deceleration: $\alpha \approx -10472.98 \text{ rad/s}^2$.
- (d) Magnitude of total acceleration: $a \approx 395168.89 \text{ m/s}^2$.